Time spent: 0 sec
Subject:
Physics

QID: 403591
System:
Mechanics and Energy

Vehicle crash testing is used to study the physics of passenger injuries and vehicle safety systems. Crash testing can simulate head-on collisions, impacts to the side of a vehicle, rollover crashes, and other situations. Likewise, crash testing can evaluate the efficacy of a range of safety devices, such as seat belts, airbags, and automatic braking systems. An automatic braking system uses sensors to detect stationary obstructions in front of the car. Based on the vehicle's speed and the distance to the object, the system automatically engages the car's brakes to avoid the collision or at least reduce the potential for injuries.

In a crash test, a vehicle is pulled toward a concrete barrier using a steel tow cable, as shown in Figure 1. The barrier is designed so most of the energy of the collision is transferred to the vehicle. Figure 2 shows the vehicle speed over time as it is accelerated to the desired speed and strikes the barrier.

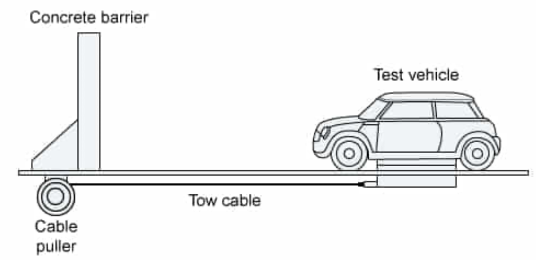


Figure 1 Schematic of a crash test setup

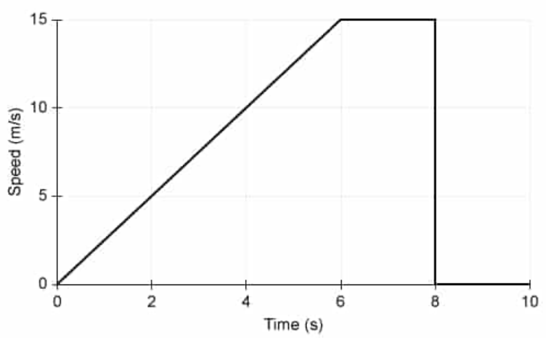


Figure 2 Vehicle speed during a crash test

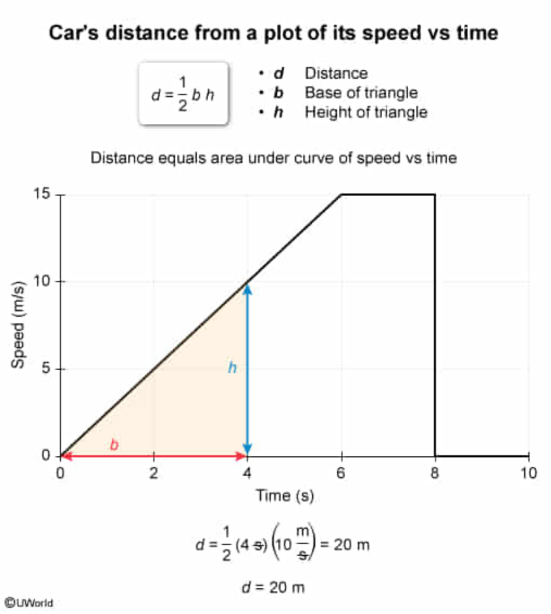
Crash test dummies are used to determine which injuries could occur to the vehicle's passengers. Various sensors within the dummies record important data, such as body acceleration, deformation of the chest cavity, and forces applied to limb bones. A dummy may contain 50 different sensors and collect 30,000 individual data points during a crash lasting 0.1 s.

In Figure 2, what is the distance traveled by the vehicle as it accelerates from 0 m/s to 10 m/s?

- A. 75 m [29%]
- B. 40 m [20%]
- ✔ C. 20 m [46%]
- D. 10 m [3%]

Correct
45% answered correctly

Explanation:



The distance *d* an object travels is equal to the product of the object's **speed** *v* and the **time** of travel *t*:

$$d = \mathrm{vt}$$

When *v* of the object is not constant, *d* can be calculated from the **area under the curve** of a plot of *v* as a function of *t*.

In this question, *v* increases linearly from 0 m/s to 10 m/s in the interval from 0 s to 4 s (ie, *v* is not constant). The area under the curve equals the area of a triangle with a base *b* of 4 s and a height *h* of 10 m/s:

$$d = \frac{1}{2}\mathrm{bh} = \frac{1}{2}(4\ \cancel{\mathrm{s}})\left(10\ \frac{\mathrm{m}}{\cancel{\mathrm{s}}}\right) = \frac{40\ \mathrm{m}}{2} = 20\ \mathrm{m}$$

Hence, the car travels 20 m as it accelerates from 0 m/s to 10 m/s.

(Choice A) 75 m is incorrectly obtained by calculating the area under the entire curve. The question asks for the distance as the car's velocity increases from 0 m/s to 10 m/s, which is only from 0 s to 4 s.

(Choice B) 40 m incorrectly results from using the formula for the area of a rectangle. The car's speed increases linearly over time, so the formula for the area of a triangle must be used.

(Choice D) 10 m is obtained by incorrectly assuming the final speed of the car, 10 m/s, is the same as the distance that the car travels.

Educational objective:
A plot of an object's speed over time can be used to determine the distance traveled by calculating the area under the curve. If the speed linearly increases during the interval of interest, the distance equals the area of the triangular region under the curve.

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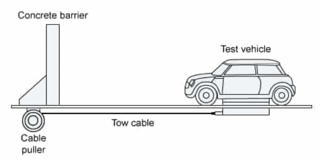

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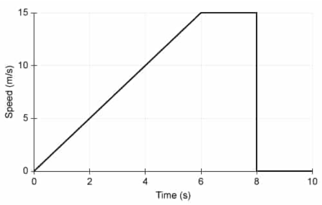


Figure 2 Vehicle speed during a crash test

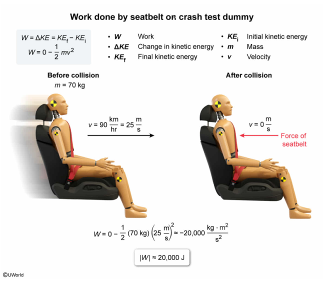
Crash test dummies are used to determine which injuries could occur to the vehicle's passengers. Various sensors within the dummies record important data, such as body acceleration, deformation of the chest cavity, and forces applied to limb bones. A dummy may contain 50 different sensors and collect 30,000 individual data points during a crash lasting 0.1 s.

A seat belt holds a 70 kg crash test dummy in the car seat during a collision that slows the car from 90 km/hr to 0 km/hr. What is the approximate magnitude of the work done by the seat belt on the crash test dummy?

- A. 2,000 J [9%]
- ✔ B. 20,000 J [44%]
- C. 40,000 J [29%]
- D. 300,000 J [17%]

Correct
44% answered correctly

Explanation:



The **work-energy theorem** states that work W done by a force on an object can be calculated as the change in the object's kinetic energy ΔKE. ΔKE is equal to the difference between the final kinetic energy KE_f and the initial kinetic energy KE_i:

$$W = \Delta \mathrm{KE} = KE_f - KE_i$$

In this question, the seat belt performs work on the crash test dummy during the collision. Before the collision, KE_i of the dummy equals half the product of the dummy's **mass** m and the square of its **velocity** v. After the collision, KE_f of the dummy equals zero because v is zero after the car hits the barrier. Therefore, W performed on the dummy by the seat belt is given by:

$$W = KE_f - KE_i = 0 - \frac{1}{2}mv^2$$

$$W = -\frac{1}{2}(70\ \mathrm{kg})\left(90\ \frac{\mathrm{km}}{\mathrm{hr}}\right)^2$$

Converting the units for v from km/hr to m/s using the relations 1 km = 1,000 m and 1 hr = 3,600 s yields:

$$v = \left(90\ \frac{\cancel{\mathrm{km}}}{\cancel{\mathrm{hr}}}\right)\left(\frac{1{,}000\ \mathrm{m}}{\cancel{\mathrm{km}}}\right)\left(\frac{\cancel{\mathrm{hr}}}{3{,}600\ \mathrm{s}}\right) = 25\ \frac{\mathrm{m}}{\mathrm{s}}$$

Therefore, W performed by the seat belt is:

$$W = -\frac{1}{2}(70\ \mathrm{kg})\left(25\ \frac{\mathrm{m}}{\mathrm{s}}\right)^2 = -\frac{1}{2}(70\ \mathrm{kg})\left(625\ \frac{\mathrm{m}^2}{\mathrm{s}^2}\right)$$

$$W = -21{,}875\ \frac{\mathrm{kg} \cdot \mathrm{m}^2}{\mathrm{s}^2} \approx -20{,}000\ \mathrm{J}$$

$$|W| \approx 20{,}000 \text{ J}$$

Hence, the magnitude of *W* performed by the seat belt is approximately equal to 20,000 J.

(Choice A) 2,000 J results from not squaring the velocity and not including the $\frac{1}{2}$ term in the kinetic energy equation.

(Choice C) 40,000 J is obtained by not including the $\frac{1}{2}$ term in the kinetic energy equation.

(Choice D) 300,000 J is incorrectly calculated by using 90 km/hr for velocity in the kinetic energy equation instead of converting the velocity to 25 m/s.

Educational objective:
The work-energy theorem states that the work performed equals the change in kinetic energy. For a vehicle collision, the work can be calculated from the change in the kinetic energy before and after the collision.

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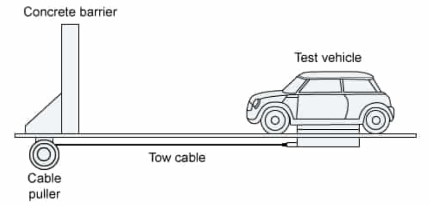

Figure 1 Schematic of a crash test setup

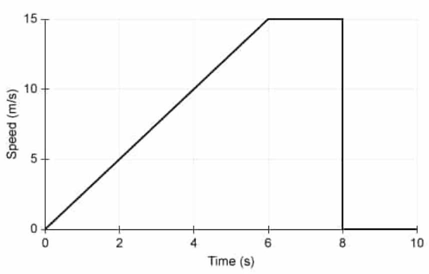

Figure 2 Vehicle speed during a crash test

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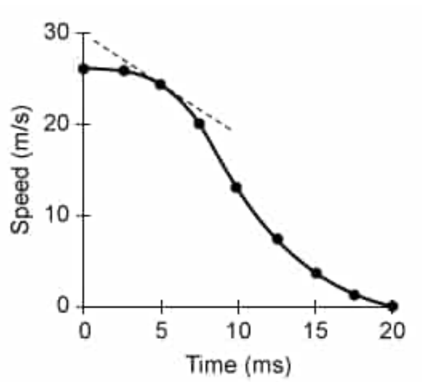
Selected data points from a speed sensor in a crash test dummy's head during a collision are plotted in the figure shown. What does the slope of the plot, denoted by the dashed line, represent?



- ✔ A. Acceleration of the head [86%]
- B. Displacement of the head [10%]
- C. Kinetic energy of the head [1%]

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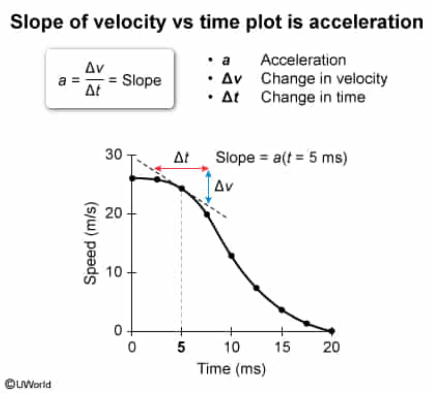
D. Work done on the head [1%]

Correct
86% answered correctly

Explanation:



Acceleration *a* is defined as the ratio of the change in **velocity** Δ*v* to the change in **time** Δ*t*:

$$a = \frac{\Delta v}{\Delta t}$$

Furthermore, the slope of a line is equal to the ratio of the rise (ie, the change along the *y*-axis) to the run (ie, the change along the *x*-axis):

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\Delta y}{\Delta x}$$

Hence, *a* equals the slope of a plot of *v* as a function of *t*.

When the plot of *v* versus *t* is not a simple straight line, the slope of the graph at each point in time is the same as the slope of the tangent line at that point. This instantaneous slope equals the **instantaneous acceleration**.

In this question, data representing *v* of the dummy's head is plotted against *t*. The slope at each time point corresponds to the slope of the tangent line, shown as the dashed line in the figure. The slope of the tangent line is equal to the ratio of Δ*v* and Δ*t*, which is the same as the instantaneous *a* of the head at 5 ms:

$$\text{Slope} = \frac{\Delta v}{\Delta t} = a$$

(Choice B) The displacement of the head equals the area under the curve of a plot of velocity versus time, not the slope of the tangent line.

(Choice C) The kinetic energy of the head depends on the velocity at each time point on the plot, but it is not equal to the slope of the tangent line.

(Choice D) The work done on the head depends on the change in kinetic energy during the collision, not the slope of the tangent line.

Educational objective:
The change in velocity of an object over time equals the acceleration of the object. The instantaneous acceleration can be calculated from the slope of the tangent line in a plot of velocity versus time.

Time spent:0 sec
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Physics

QID:403594
System:
Mechanics and Energy